```
devasc@labvm:~/labs/devnet-src/python$ nano circleClass.py
devasc@labvm:~/labs/devnet-src/python$ cd ~/labs/devnet-src
devasc@labvm:~/labs/devnet-src$ mkdir -p unittest
devasc@labvm:~/labs/devnet-src$ cd unittest
devasc@labvm:~/labs/devnet-src/unittest$ pwd
/home/devasc/labs/devnet-src/unittest
devasc@labvm:~/labs/devnet-src/unittest$
```

Вопрос 1. Какой класс `unittest` используется для создания отдельного набора тестов?

Ответ: `unittest.TestCase`.

Вопрос 2. Как тестовый запускатель определяет, какие методы являются тестами?

Ответ: По имени метода — оно должно начинаться с `test_`.

Вопрос 3. Какая команда выводит список всех параметров командной строки `unittest`?

Ответ:

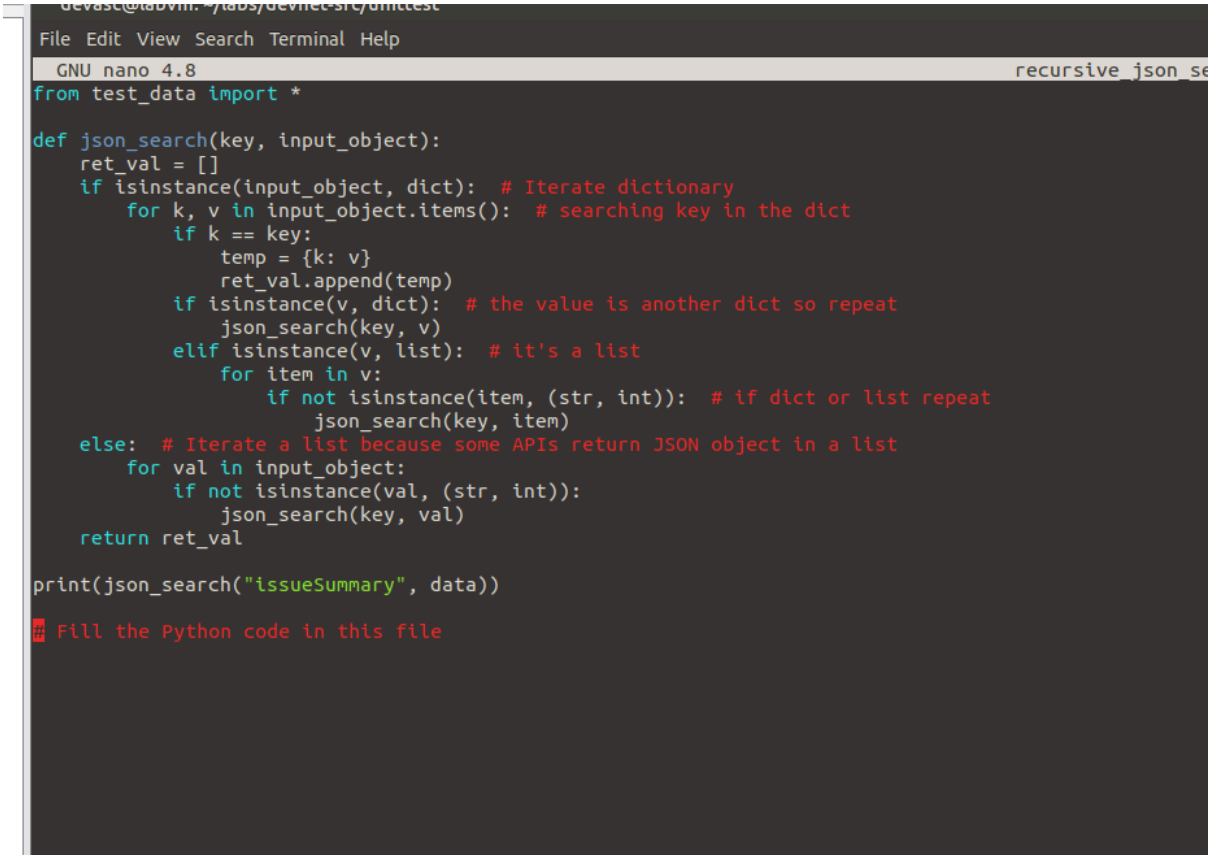
python3 -m unittest -h

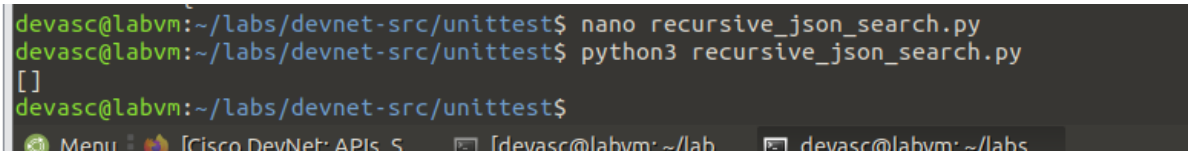
PART 3

```
devasc@labvm:~/labs/devnet-src/unittest$ cd ~/labs/devnet-src/unittest
devasc@labvm:~/labs/devnet-src/unittest$ ls -l
total 16
-rw-r--r-- 1 devasc devasc   36 Apr 23  2020 recursive_json_search.py
-rw-r--r-- 1 devasc devasc 4735 Apr 23  2020 test_data.py
-rw-r--r-- 1 devasc devasc   36 Apr 23  2020 test_json_search.py
devasc@labvm:~/labs/devnet-src/unittest$ more test_data.py
key1 = "issueSummary"
key2 = "XY&^$#*@!1234%^&"

data = {
  "id": "AWcvsjx864kVeDHDi2gB",
  "instanceId": "E-NETWORK-EVENT-AWcvsjx864kVeDHDi2gB-1542693469197",
  "category": "Warn",
  "status": "NEW",
  "timestamp": 1542693469197,
  "severity": "P1",
  "domain": "Availability",
  "source": "DNAC",
  "priority": "P1",
  "type": "Network",
  "title": "Device unreachable",
  "description": "This network device leaf2.abc.inc is unreachable from controller. The device role is ACCESS.",
  "actualServiceId": "10.10.20.82",
  "assignedTo": "",
  "enrichmentInfo": {
    "issueDetails": {
```

наши кей 1 и кей 2


File Edit View Search Terminal Help
GNU nano 4.8 recursive_json_se
from test_data import *
def json_search(key, input_object):
ret_val = []
if isinstance(input_object, dict): # Iterate dictionary
for k, v in input_object.items(): # searching key in the dict
if k == key:
temp = {k: v}
ret_val.append(temp)
if isinstance(v, dict): # the value is another dict so repeat
json_search(key, v)
elif isinstance(v, list): # it's a list
for item in v:
if not isinstance(item, (str, int)): # if dict or list repeat
json_search(key, item)
else: # Iterate a list because some APIs return JSON object in a list
for val in input_object:
if not isinstance(val, (str, int)):
json_search(key, val)
return ret_val
print(json_search("issueSummary", data))
Fill the Python code in this file


devasc@labvm:~/labs/devnet-src/unittest$ nano recursive_json_search.py
devasc@labvm:~/labs/devnet-src/unittest$ python3 recursive_json_search.py
[]
devasc@labvm:~/labs/devnet-src/unittest$

```
devasc@labvm: ~/labs/devnet-src/unittest
File  Edit  View  Search  Terminal  Help
  GNU nano 4.8                                                    test_json_search
import unittest
from recursive_json_search import *
from test_data import *

class json_search_test(unittest.TestCase):
    '''test module to test search function in `recursive_json_search.py`'''

    def test_search_found(self):
        '''key should be found, return list should not be empty'''
        self.assertTrue([] != json_search(key1, data))

    def test_search_not_found(self):
        '''key should not be found, should return an empty list'''
        self.assertTrue([] == json_search(key2, data))

    def test_is_a_list(self):
        '''Should return a list'''
        self.assertIsInstance(json_search(key1, data), list)

if __name__ == '__main__':
    unittest.main()

# Fill the Python code in this file
```

```
[]
devasc@labvm:~/labs/devnet-src/unittest$ nano test_json_search.py
devasc@labvm:~/labs/devnet-src/unittest$ python3 test_json_search.py
[]
.F.
======================================================================
FAIL: test_search_found (__main__.json_search_test)
key should be found, return list should not be empty
----------------------------------------------------------------------
Traceback (most recent call last):
  File "test_json_search.py", line 10, in test_search_found
    self.assertTrue([] != json_search(key1, data))
AssertionError: False is not true

----------------------------------------------------------------------
Ran 3 tests in 0.001s

FAILED (failures=1)
devasc@labvm:~/labs/devnet-src/unittest$
Menu  [Cisco DevNet: APIs, S...  [devasc@labvm: ~/lab...  devasc@labvm: ~/labs...
```

stept 5

devasc@labvm: ~/labs/devnet-src/unittest

File Edit View Search Terminal Help

GNU nano 4.8 recursive_json_search.py

```
from test_data import *

ret_val = []

def json_search(key, input_object):
    if isinstance(input_object, dict):  # Iterate dictionary
        for k, v in input_object.items():  # searching key in the dict
            if k == key:
                temp = {k: v}
                ret_val.append(temp)
            if isinstance(v, dict):  # the value is another dict so repeat
                json_search(key, v)
            elif isinstance(v, list):  # it's a list
                for item in v:
                    if not isinstance(item, (str, int)):  # if dict or list repeat
                        json_search(key, item)
    else:  # Iterate a list because some APIs return JSON object in a list
        for val in input_object:
            if not isinstance(val, (str, int)):
                json_search(key, val)
    return ret_val

print(json_search("issueSummary", data))

# Fill the Python code in this file
```

```
devasc@labvm:~/labs/devnet-src/unittest$ nano recursive_json_search.py
devasc@labvm:~/labs/devnet-src/unittest$ python3 recursive_json_search.py
[{'issueSummary': 'Network Device 10.10.20.82 Is Unreachable From Controller'}]
devasc@labvm:~/labs/devnet-src/unittest$
```

```
devasc@labvm:~/labs/devnet-src/unittest$ python3 recursive_json_search.py
[{'issueSummary': 'Network Device 10.10.20.82 Is Unreachable From Controller'}]
devasc@labvm:~/labs/devnet-src/unittest$ python3 -m unittest test_json_search
[{'issueSummary': 'Network Device 10.10.20.82 Is Unreachable From Controller'}]
..F
======================================================================
FAIL: test_search_not_found (test_json_search.json_search_test)
key should not be found, should return an empty list
----------------------------------------------------------------------
Traceback (most recent call last):
  File "/home/devasc/labs/devnet-src/unittest/test_json_search.py", line 14, in test_search_not_found
    self.assertTrue([] == json_search(key2, data))
AssertionError: False is not true

----------------------------------------------------------------------
Ran 3 tests in 0.001s

FAILED (failures=1)
devasc@labvm:~/labs/devnet-src/unittest$
```

step 6

devasc@labvm: ~/labs/devnet-src/unittest

File Edit View Search Terminal Help

GNU nano 4.8 test_data.py

```
key1 = "issueSummary"
key2 = "XY&^$#*@!1234%^&"

data = {
  "id": "AWcvsjx864kVeDHDi2gB",
  "instanceId": "E-NETWORK-EVENT-AWcvsjx864kVeDHDi2gB-1542693469197",
  "category": "Warn",
  "status": "NEW",
  "timestamp": 1542693469197,
  "severity": "P1",
  "domain": "Availability",
  "source": "DNAC",
  "priority": "P1",
  "type": "Network",
  "title": "Device unreachable",
  "description": "This network device leaf2.abc.inc is unreachable from controller. The device role is ACCESS.",
  "actualServiceId": "10.10.20.82",
  "assignedTo": "",
  "enrichmentInfo": {
    "issueDetails": {
      "issue": [
        {
          "issueId": "AWcvsjx864kVeDHDi2gB",
          "issueSource": "Cisco DNA",
          "issueCategory": "Availability",
          "issueName": "snmp_device_down",
          "issueDescription": "This network device leaf2.abc.inc is unreachable from controller. The device role is ACCESS.",
          "issueEntity": "network_device",
          "issueEntityValue": "10.10.20.82",
          "issueSeverity": "HIGH",
          "issuePriority": "",
          "issueSummary": "Network Device 10.10.20.82 Is Unreachable From Controller",
          "issueTimestamp": 1542693469197,
          "suggestedActions": [
            {
              "message": "From the controller, verify whether the last hop is reachable.",
              "steps": []
            },
            {
              "message": "Verify that the physical port(s) on the network device associated with the network device discovery(IP) is UP.",
```

[Read 116 lines]

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo M-A Mark Text M-] To Bracket M-Q

devasc@labvm: ~/labs/devnet-src/unittest

File Edit View Search Terminal Help

GNU nano 4.8 recursive_json_search.py

```
from test_data import *

def json_search(key, input_object):
    """
    Search a key from JSON object, get nothing back if key is not found
    key : "keyword" to be searched, case sensitive
    input_object : JSON object to be parsed, test_data.py in this case
    inner_function() is actually doing the recursive search
    return a list of key:value pair
    """
    ret_val = []

    def inner_function(key, input_object):
        if isinstance(input_object, dict):
            for k, v in input_object.items():
                if k == key:
                    temp = {k: v}
                    ret_val.append(temp)

                if isinstance(v, dict):
                    inner_function(key, v)
                elif isinstance(v, list):
                    for item in v:
                        if not isinstance(item, (str, int)):
                            inner_function(key, item)
        else:
            for val in input_object:
                if not isinstance(val, (str, int)):
                    inner_function(key, val)

    inner_function(key, input_object)
    return ret_val


print(json_search("issueSummary", data))
# Fill the Python code in this file
```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo M-A Mark

```
devasc@labvm:~/labs/devnet-src/unittest$ nano test_data.py
devasc@labvm:~/labs/devnet-src/unittest$ nano recursive_json_search.py
devasc@labvm:~/labs/devnet-src/unittest$ python3 -m unittest
[{'issueSummary': 'Network Device 10.10.20.82 Is Unreachable From Controller'}]
...
----------------------------------------------------------------------
Ran 3 tests in 0.000s

OK
devasc@labvm:~/labs/devnet-src/unittest$ python3 -m unittest test_json_search
[{'issueSummary': 'Network Device 10.10.20.82 Is Unreachable From Controller'}]
...
----------------------------------------------------------------------
Ran 3 tests in 0.000s

OK
devasc@labvm:~/labs/devnet-src/unittest$
```